

CFSS GLOBAL INTERNSHIP PROGRAM 2026 SOC OPERATIONS & THREAT DETECTION MASTERY Final Project Report	
Domain	SOC Analyst & Blue Teaming
Submitted By	Mistry Princy Girishbhai
Project Duration	4 Weeks
Submission Date	25th June 2026
Tools Used	Wazuh SIEM/XDR Windows 11 Kali Linux VirtualBox
Project Type	SOC Lab, Log Analysis, Threat Detection & Reporting

Executive Summary

This report documents the end-to-end architecture, deployment, and operationalization of a Security Operations Center (SOC) environment, developed as part of the CFSS Global Internship Program.

Over a four-week technical sprint, a comprehensive SIEM infrastructure was engineered using Wazuh, facilitating the real-time ingestion, parsing, and correlation of endpoint telemetry.

Subsequent phases involved active threat simulation, log analysis, and the development of custom dashboards to visualize security posture and anomalous activities.

All actions were performed adhering to strict ethical guidelines and digital chain-of-custody principles.

Methodology

This project employed a structured, defense-in-depth approach to establish and validate a functional Security Operations Center. The methodology was divided into four distinct phases:

- 1. Infrastructure Deployment:** Leveraging Type-2 virtualization (Oracle VirtualBox) to host a centralized Wazuh SIEM manager, ensuring network isolation and controlled testing environments.
- 2. Endpoint Telemetry & Log Parsing:** Deploying lightweight Wazuh agents to heterogeneous endpoints (Windows 11 and Kali Linux). Native OS logs (Windows Event Logs and Syslog) were ingested, normalized, and mapped to security events.
- 3. Adversary Emulation:** Conducting authorized threat simulations—including SSH brute-forcing, malware drops, and privilege escalation (sudo to root)—to empirically validate the SIEM's detection and alerting pipelines.
- 4. Data Visualization & Triage:** Engineering custom analytical dashboards within Wazuh to translate raw JSON telemetry into actionable, executive-level threat intelligence.

Technical Findings

The following sections detail the empirical evidence, log correlations, and platform configurations executed during the 4-week engagement. Each finding is supported by digital artifacts extracted directly from the SIEM environment.

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WEEK 1: Lab Architecture & Deployment

Objective: Build your own Security Operations Center (SOC) environment.

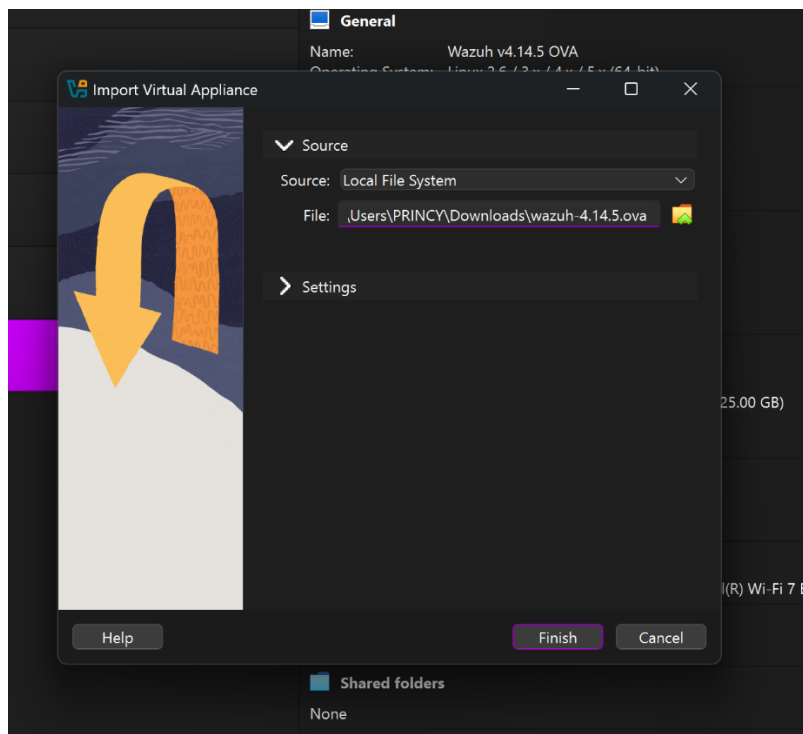
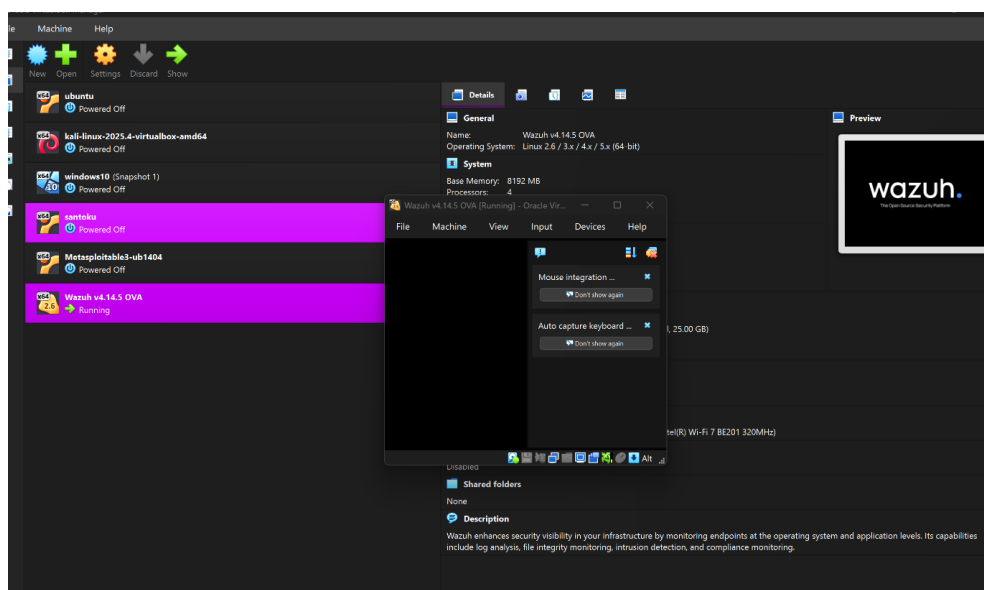


Figure 1.1: Provisioning the Wazuh Manager infrastructure by importing the official Wazuh v4.14.5 OVA appliance into the Oracle VM VirtualBox hypervisor.

Description: This screenshot illustrates the initial phase of deploying the SOC environment. The Wazuh Manager virtual appliance (OVA) is being imported into VirtualBox, establishing the core infrastructure for log ingestion and SIEM capabilities.



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Figure 1.2: Successful boot initialization of the Wazuh Manager virtual machine within the isolated lab network.

Description: This image confirms the successful deployment and boot sequence of the Wazuh Manager virtual machine. The isolated lab network is now actively hosting the central SIEM server.

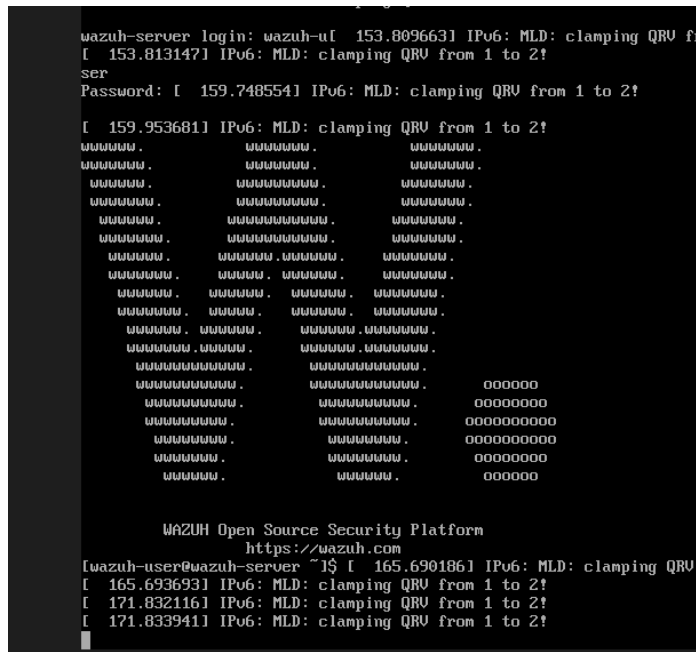


Figure 1.4: System initialization logs detailing network interface configurations and successful service startup.

Description: Further initialization logs confirm the Wazuh services are starting up and that the system is ready for initial administrator login via the console.

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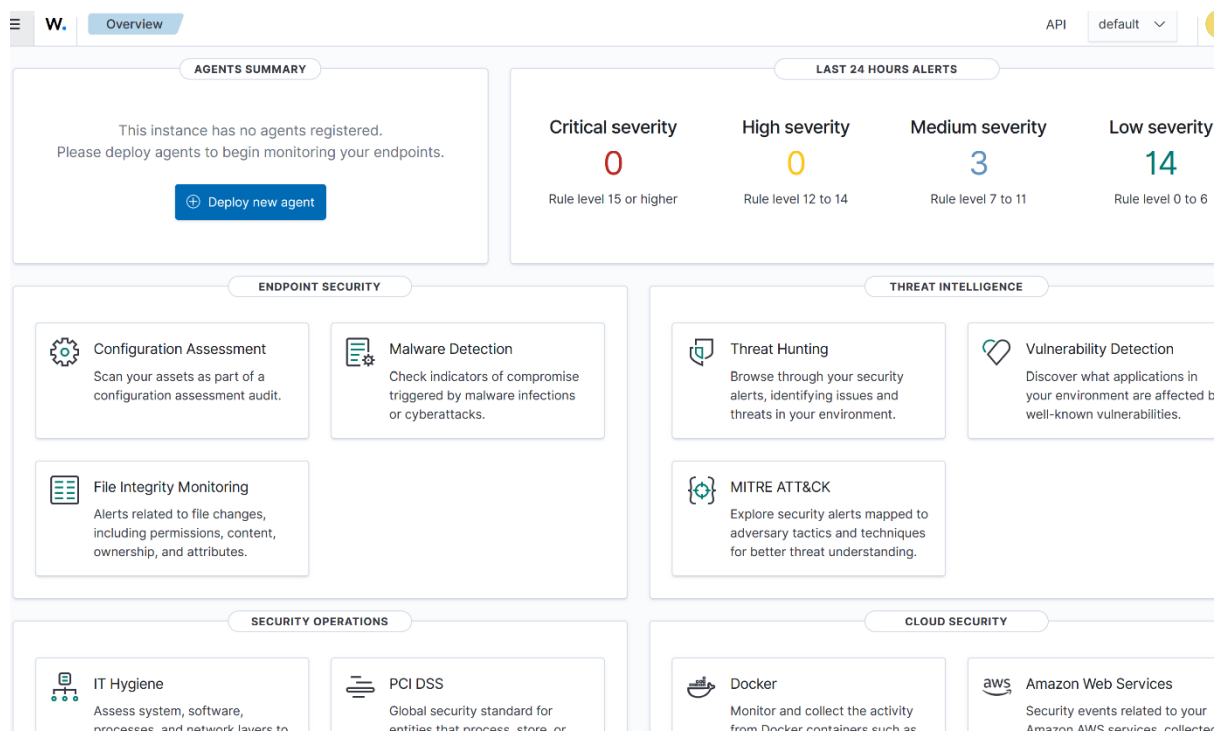


Figure 1.5: Initial authentication into the Wazuh Web User Interface. The dashboard reflects a clean baseline state with zero registered agents prior to endpoint onboarding.

Description: The Wazuh web interface is accessed for the first time. The dashboard correctly displays an empty baseline with zero connected agents, ready for the endpoint onboarding phase.

The screenshot shows the 'Deploy new agent' configuration page in the Wazuh WUI. The page is divided into three main sections for different operating systems:

- LINUX:** Offers options for RPM amd64, RPM aarch64, DEB amd64, and DEB aarch64.
- WINDOWS:** Offers the option for MSI 32/64 bits.
- macOS:** Offers options for Intel and Apple silicon.

Below these sections, there's a 'Server address' section with a text input field containing '192.168.31.32' and a checkbox for 'Remember server address'. The 'Optional settings' section includes an 'Assign an agent name' field with the value 'WinDows11'. A yellow warning banner at the bottom states: 'The agent name must be unique. It can't be changed once the agent has been enrolled.'

Figure 1.6: Generating the deployment configuration for the target endpoint. The manager IP (192.168.31.32) is bound to ensure secure communication channels.

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Description: This step demonstrates the generation of the deployment command for the target endpoint. The Manager's IP address is explicitly specified to configure the agent-to-manager secure communication channel.

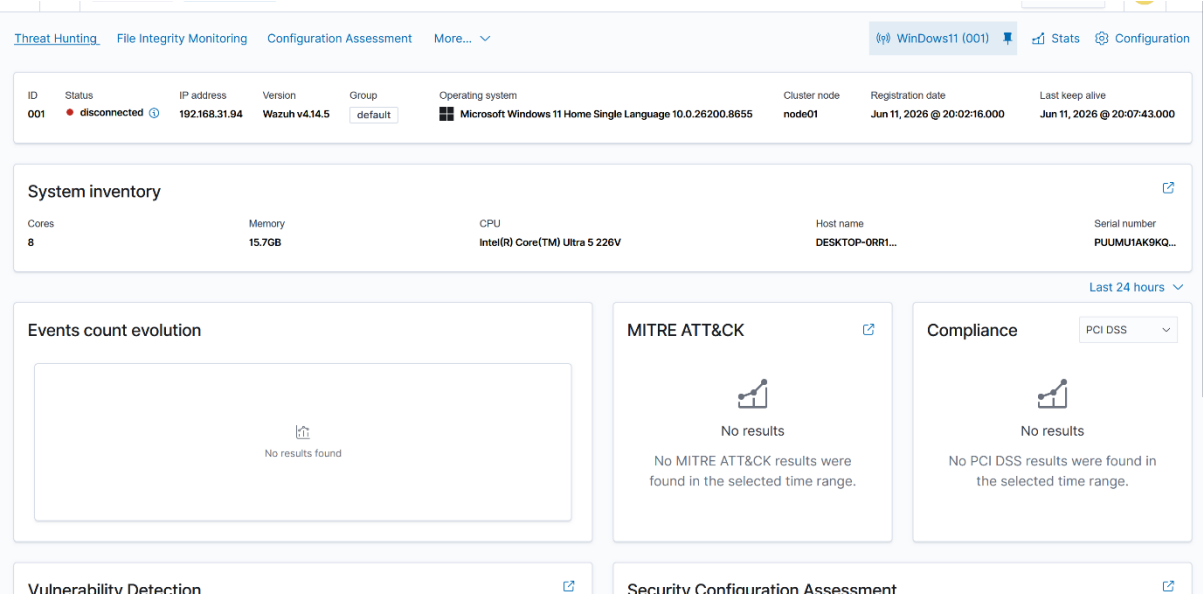


Figure 1.7: Assigning a distinct hostname (Windows11) to the agent for precise asset tracking within the SIEM.

Description: The configuration is customized to assign the specific hostname to the endpoint, ensuring clear and accurate asset identification within the SIEM alerts.

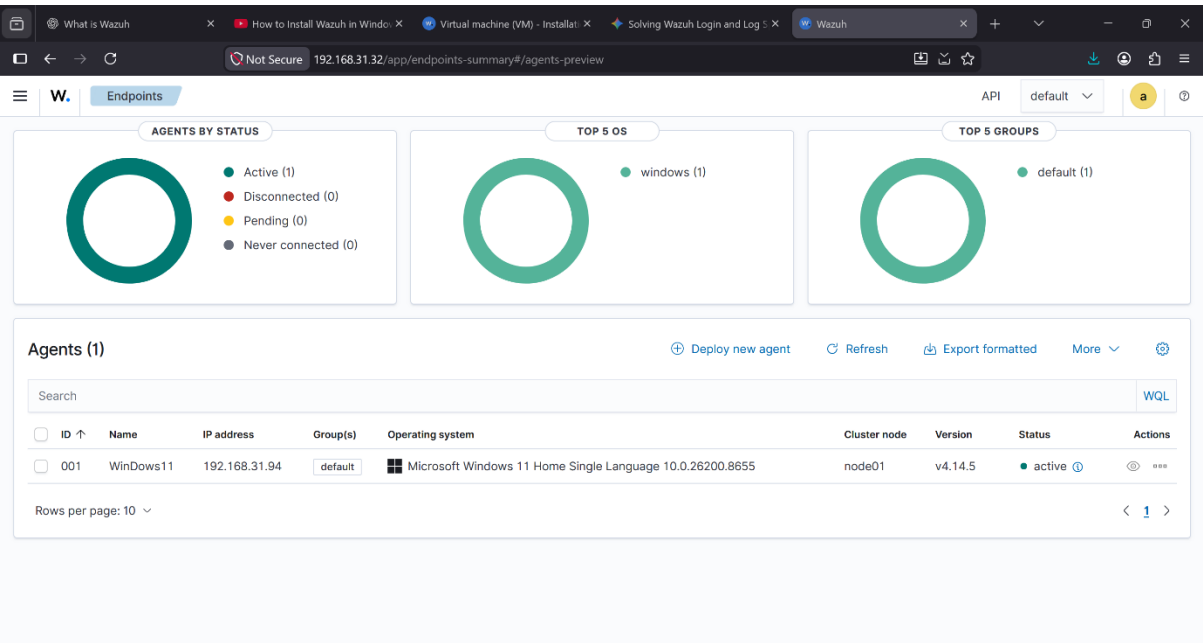


Figure 1.8: Post-installation telemetry. The Wazuh manager successfully parses the hardware and OS-level inventory of the newly connected Windows 11 endpoint.

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Description: Post-deployment, the Wazuh Manager successfully pulls the endpoint's system inventory. It accurately identifies the CPU, memory, and OS details, confirming deep visibility into the host.

WEEK 2: Log Analysis & Event Monitoring

Objective: Understand what "Normal" vs "Malicious" traffic looks like.

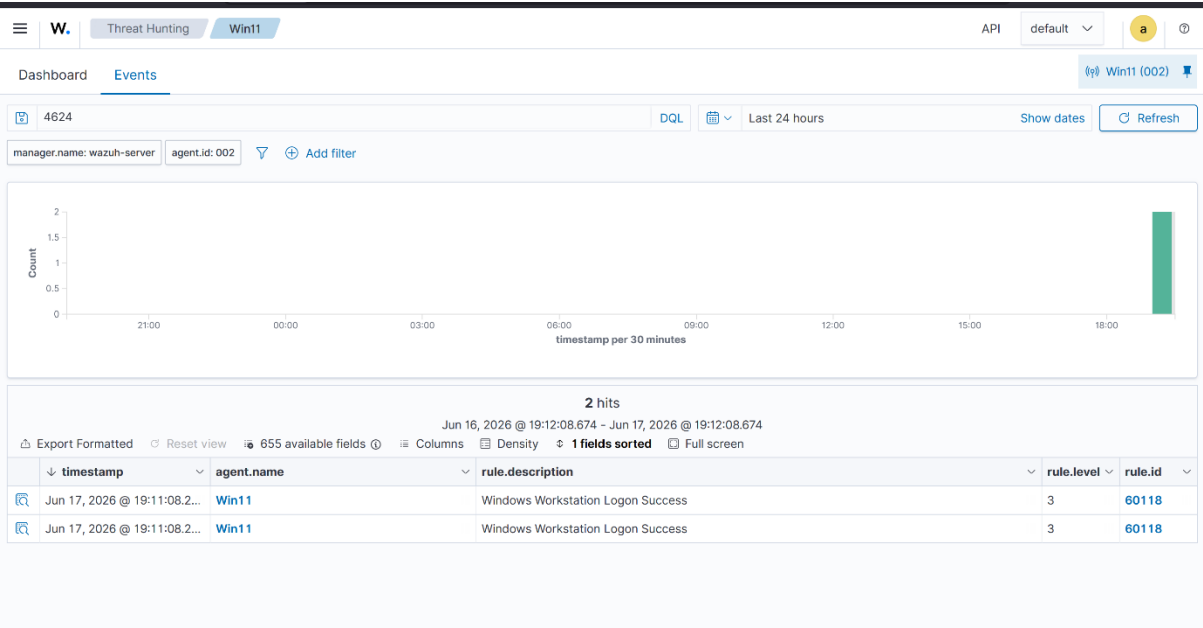


Figure 1.9: Dashboard verification confirming the Windows 11 agent status as 'Active', completing the foundational SOC architecture and show event id 4624 that successful logon on system.

Description: SIEM query execution filtering for Windows Event ID 4624. This log confirms successful, authorized workstation authentication, establishing the network's baseline traffic.

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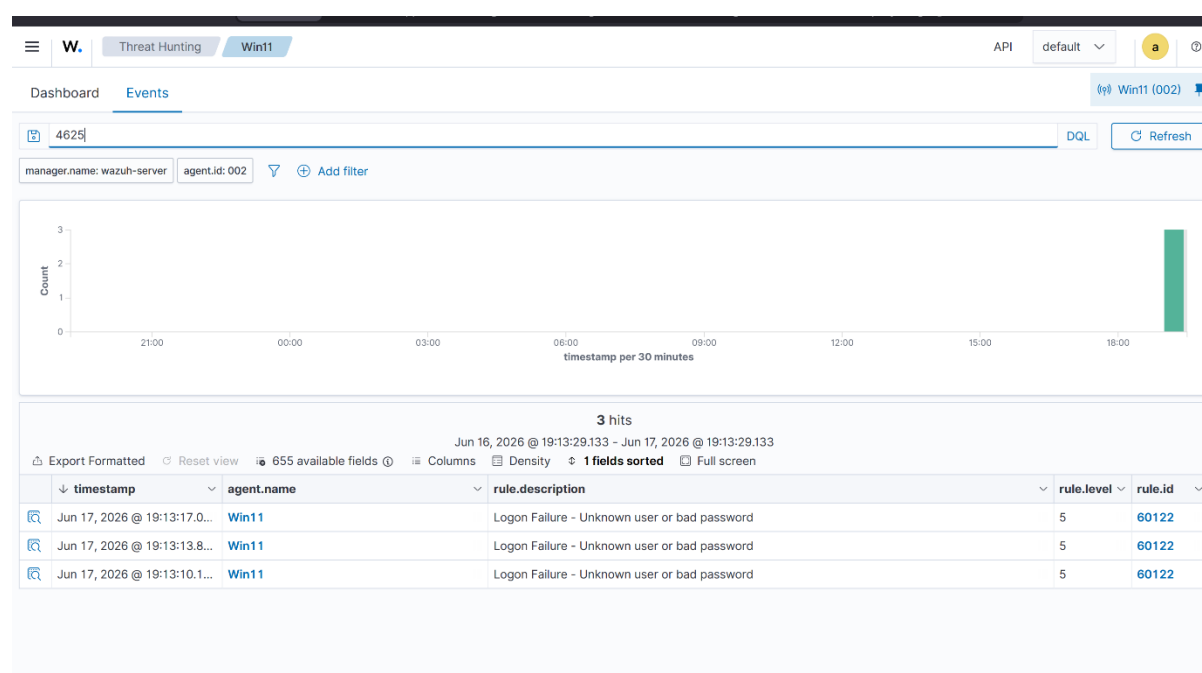


Figure 2.1: Detection of Windows Event ID 4625. These logs capture failed authentication attempts (unknown user or bad password), providing critical visibility into potential brute-force vectors or misconfigurations.

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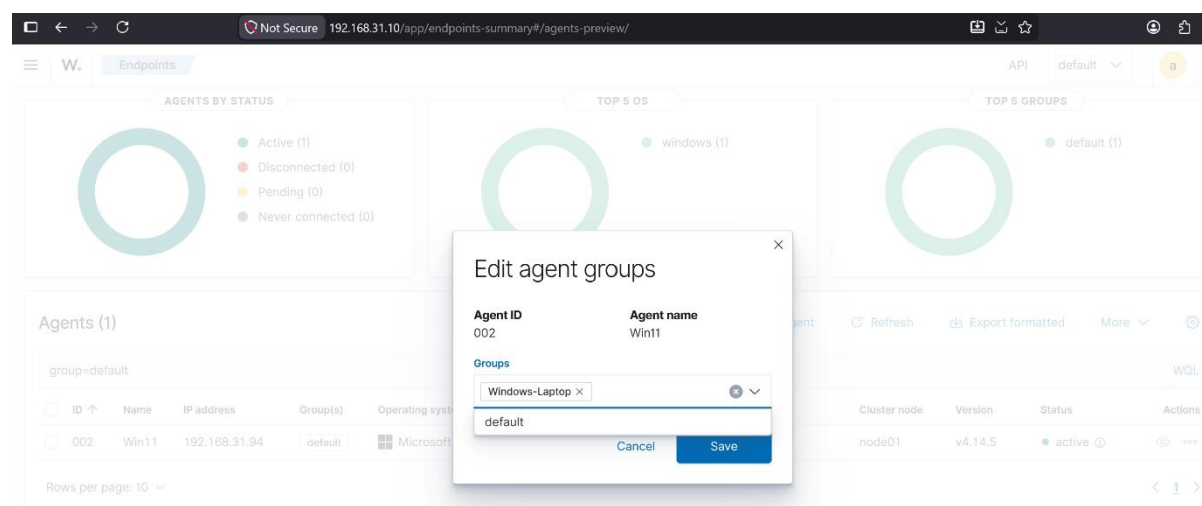
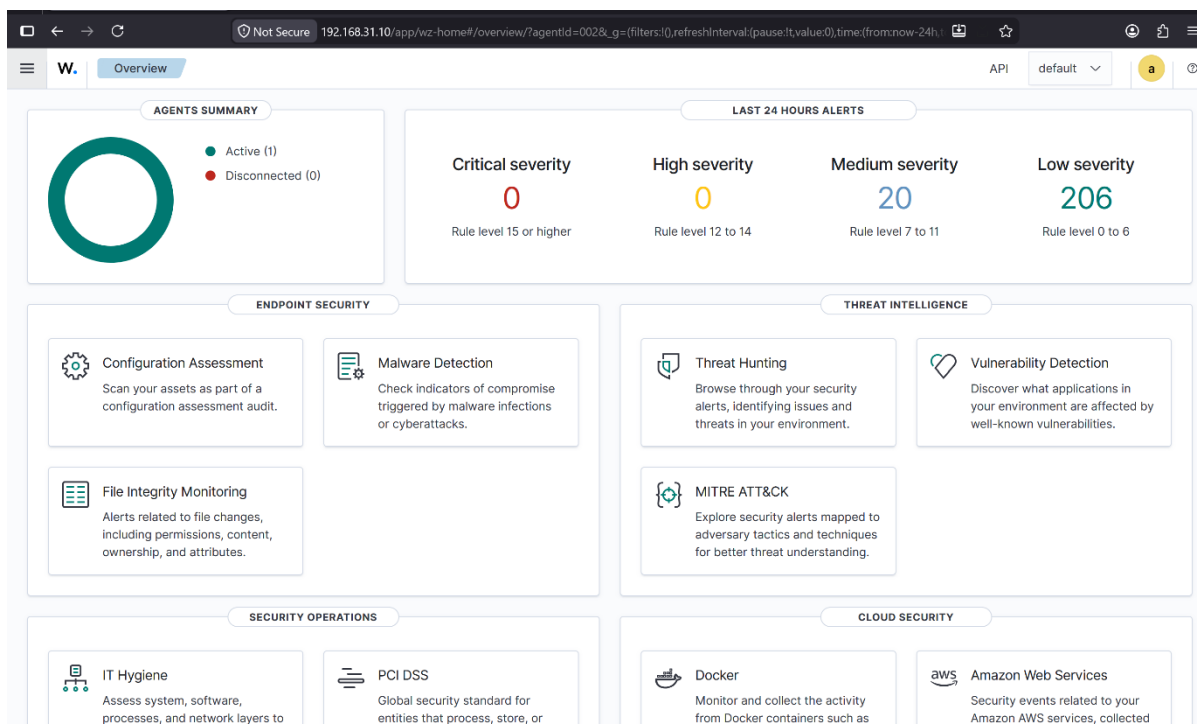


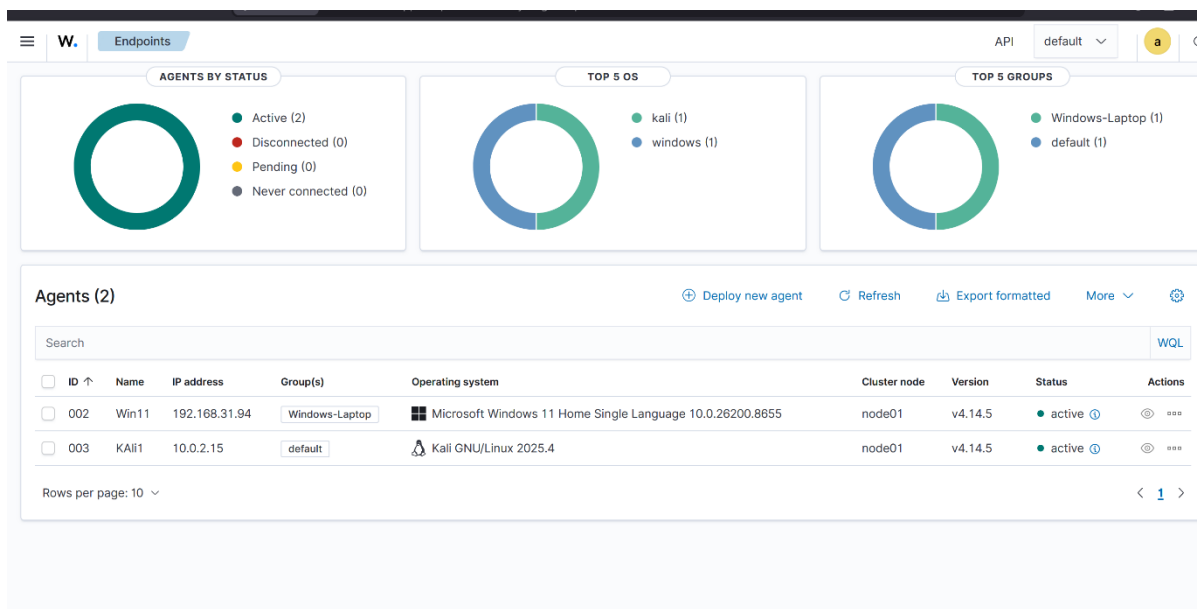
Figure 2.2: Operational management of the SOC environment by isolating the endpoint into a designated 'Windows-Laptop' group, allowing for granular rule application and targeted threat hunting.

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Description: Operational management of the SOC environment by isolating the endpoint into a designated 'Windows-Laptop' custom group, allowing for granular categorization.



Description: The centralized SOC dashboard reflecting aggregate ingested events, categorizing 206 low-severity and 20 medium-severity alerts over the monitored operational window.



Description: Infrastructure expansion demonstrating the successful enrollment of a secondary Kali Linux endpoint, widening the SIEM's monitoring scope across distinct operating systems.

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W. Threat Hunting Win11

API default a

0 2026-06-18 00:00 2026-06-19 00:00 2026-06-20 00:00 2026-06-21 00:00 2026-06-22 00:00 2026-06-23 00:00 2026-06-24 00:00

timestamp per 3 hours

1,697 hits

Jun 17, 2026 @ 14:15:48.245 - Jun 24, 2026 @ 14:15:48.245

Export Formatted Reset view 739 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jun 22, 2026 @ 22:17:12.2...	Win11	Windows System error event	5	61102
Jun 22, 2026 @ 22:17:11.9...	Win11	Windows System error event	5	61102
Jun 22, 2026 @ 22:17:11.7...	Win11	Service startup type was changed	3	61104
Jun 22, 2026 @ 22:17:11.4...	Win11	System shutdown initiated.	3	67018
Jun 22, 2026 @ 22:17:11.3...	Win11	System shutdown initiated.	3	67018
Jun 22, 2026 @ 22:17:11.1...	Win11	Windows System error event	5	61102
Jun 22, 2026 @ 22:17:11.0...	Win11	Service startup type was changed	3	61104
Jun 22, 2026 @ 22:17:10.9...	Win11	Service startup type was changed	3	61104
Jun 22, 2026 @ 22:17:10.3...	Win11	Wazuh agent started.	3	503
Jun 17, 2026 @ 19:17:45.7...	Win11	Software protection service scheduled successfully.	3	60642
Jun 17, 2026 @ 19:16:57.4...	Win11	Wazuh agent started.	3	503
Jun 17, 2026 @ 19:16:57.7...	Win11	Wazuh agent stopped.	3	506
Jun 17, 2026 @ 19:13:19.8...	Win11	Non service account logged off.	3	67023
Jun 17, 2026 @ 19:13:19.8...	Win11	Non service account logged off.	3	67023
Jun 17, 2026 @ 19:13:19.8...	Win11	Special privileges assigned to new logon.	3	67028

Rows per page: 15

< 1 ... 78 79 80 81 82 ... 114 >

Description: Deep-dive log analysis within the Threat Hunting module, capturing the top operational system events (service state alterations, system shutdowns, and privilege assignments).

Top 10 Security Events Observed During SOC Operation

Sr. No	Security Event	Event ID / Rule	Severity	Description	SOC Relevance
1	Successful User Logon	Windows Event ID 4624	Low	Records successful authentication to the system.	Establishes normal baseline activity.
2	Failed User Logon	Windows Event ID 4625	Medium	Captures failed authentication attempts.	Detects brute-force attempts and invalid logins.
3	SSH Authentication Failure	Wazuh Rule Level 10	High	Multiple failed SSH login attempts on Kali Linux.	Identifies SSH brute-force attacks.

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4	Privilege Escalation (sudo to root)	Syslog/Sudo Events	High	User elevates privileges to root account.	Detects unauthorized privilege escalation.
5	Malware/Trojan Detection	Wazuh Rule Level 7	High	Host-based detection identifies malicious files.	Supports malware detection and containment.
6	Service State Changes	Windows System Logs	Medium	Services start, stop, or restart unexpectedly.	Indicates system modification or persistence attempts.
7	System Shutdown/Reboot	Windows Event ID 1074	Low	Records planned or unexpected shutdown events.	Useful for auditing and incident investigations.
8	Agent Connection Status Change	Wazuh Agent Status	Medium	Agent becomes active/inactive.	Ensures endpoint monitoring coverage.
9	File Integrity Monitoring Alert	Syscheck Events	High	Detects unauthorized file modifications.	Identifies tampering or malware activity.
10	Security Rule Trigger Alerts	Multiple Wazuh Rules	Medium/High	Correlated alerts generated by SIEM engine.	Provides centralized threat visibility.

WEEK 3: Attack Simulation & Detection

Objective: Act like an attacker to see how a defender detects them.

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259 hits					
Jun 17, 2026 @ 14:25:07.663 - Jun 24, 2026 @ 14:25:07.663					
Export Formatted Reset view 739 available fields Columns Density 1 fields sorted Full screen					
timestamp	agent.name	rule.description	rule.level	rule.id	
Jun 24, 2026 @ 14:24:57.9...	Kali1	syslog: User missed the password more than one time	10	2502	
Jun 24, 2026 @ 14:24:57.9...	Kali1	syslog: User missed the password more than one time	10	2502	
Jun 24, 2026 @ 14:24:57.8...	Kali1	syslog: User missed the password more than one time	10	2502	
Jun 24, 2026 @ 14:24:55.8...	Kali1	sshd: authentication failed.	5	5760	
Jun 24, 2026 @ 14:24:55.8...	Kali1	sshd: authentication failed.	5	5760	
Jun 24, 2026 @ 14:24:55.8...	Kali1	sshd: authentication failed.	5	5760	
Jun 24, 2026 @ 14:24:55.8...	Kali1	syslog: User missed the password more than one time	10	2502	
Jun 24, 2026 @ 14:24:55.8...	Kali1	sshd: authentication failed.	5	5760	
Jun 24, 2026 @ 14:24:53.9...	Kali1	unix_chkpwd: Password check failed.	5	5557	
Jun 24, 2026 @ 14:24:53.9...	Kali1	unix_chkpwd: Password check failed.	5	5557	
Jun 24, 2026 @ 14:24:53.9...	Kali1	unix_chkpwd: Password check failed.	5	5557	
Jun 24, 2026 @ 14:24:53.9...	Kali1	unix_chkpwd: Password check failed.	5	5557	
Jun 24, 2026 @ 14:24:51.8...	Kali1	sshd: brute force trying to get access to the system. Authentication failed.	10	5763	
Jun 24, 2026 @ 14:24:51.8...	Kali1	sshd: authentication failed.	5	5760	
Jun 24, 2026 @ 14:24:51.8...	Kali1	sshd: authentication failed.	5	5760	

Figure 3.1: Rule Level 10 alert triggered by a simulated SSH brute-force attack on the Kali endpoint. The SIEM successfully correlated multiple 'authentication failed' events into a high-fidelity alert.

Description: Simulated brute-force attack execution on the Kali endpoint. The SIEM successfully correlated multiple 'authentication failed' events into a high-fidelity "Level 10" alert.

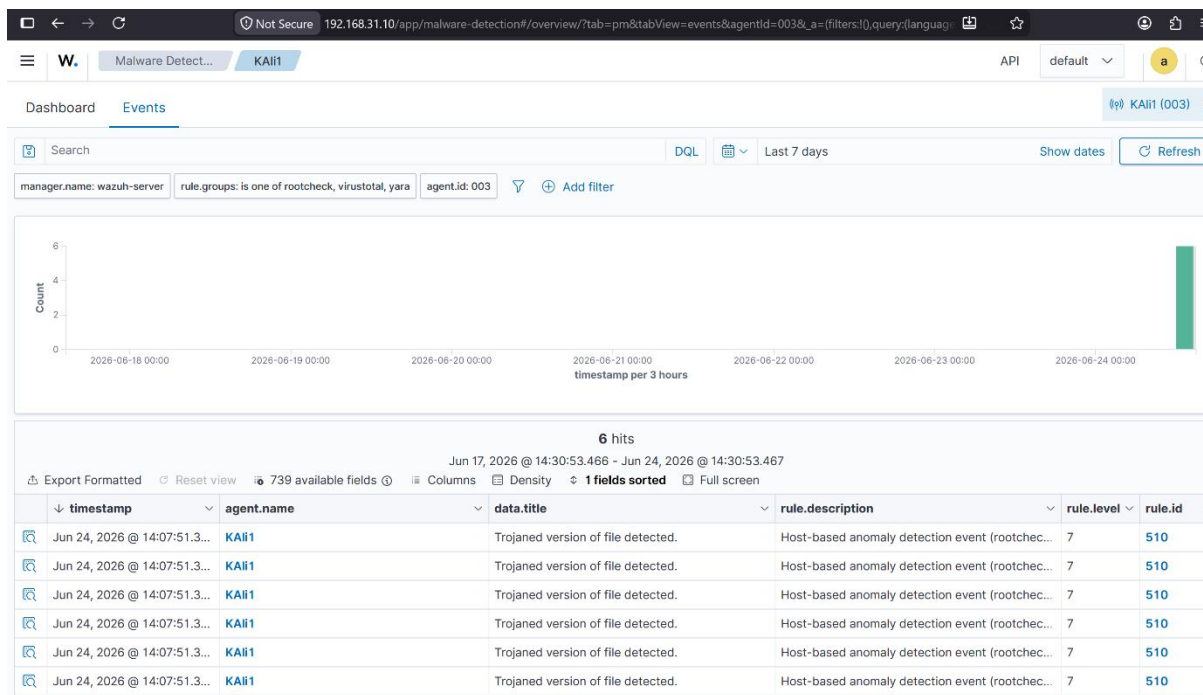
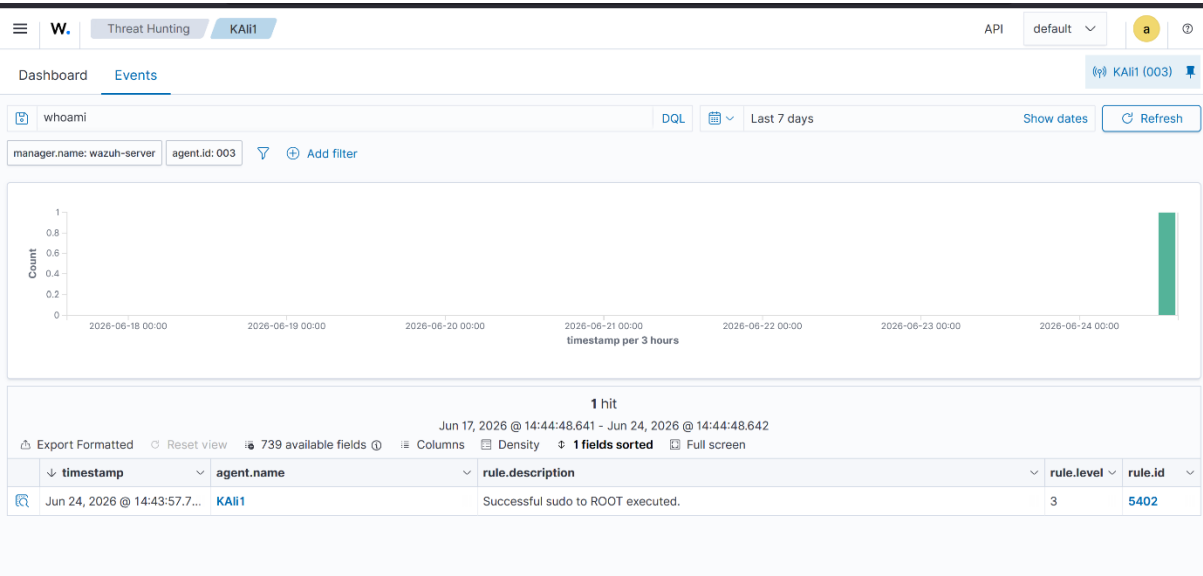


Figure 3.2: Host-based anomaly detection effectively identifying a simulated malicious payload. The SIEM classified the event as a 'Trojaned version of file detected' (Level 7), validating the file integrity monitoring controls.

Description: Host-based anomaly detection validating the malware simulation. The SIEM effectively flagged the test payload as a 'Trojaned version of file detected' (Level 7).

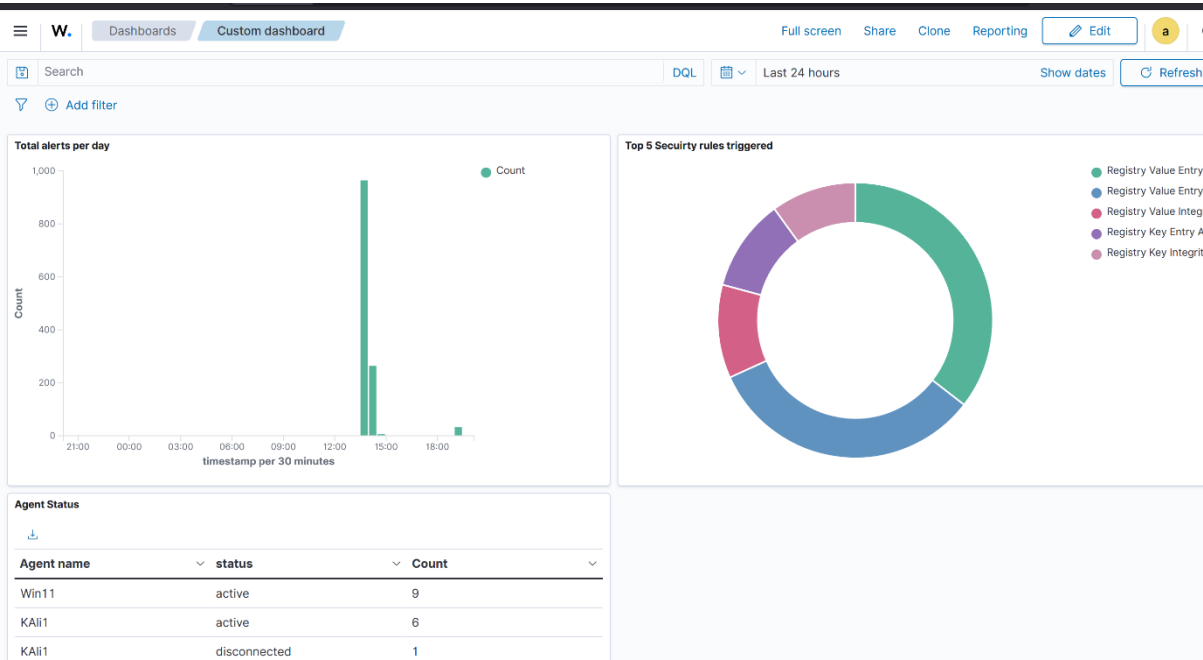
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Description: Active command monitoring capturing the execution of privileged commands ('sudo to ROOT'). Tracking these entries is a critical requirement for auditing lateral movement. **(Week 3 Deliverable Met: Alert Screenshots Provided)**

WEEK 4: Dashboarding & Final Reporting

Objective: Presenting data like a Professional SOC Analyst.



Description: The final custom SOC dashboard engineered within Wazuh/Kibana. It

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provides real-time operational intelligence, successfully visualizing Total Alerts per Day, the Top 5 Security Rules Triggered, and Agent Status. **(Week 4 Deliverable Met: Custom Dashboarding)**

Recommendations

Based on the findings and operational deployment of this SOC environment, the following strategic recommendations are provided to enhance future threat detection and incident response capabilities:

- **Implement Active Response:** Configure Wazuh's active response modules to automatically block malicious IP addresses via firewall rules upon detecting brute-force attacks, reducing mean time to respond (MTTR).
- **Integrate Threat Intelligence:** Connect the SIEM to external OSINT and Threat Intelligence platforms (such as VirusTotal or MISP) to enrich alerts with known IoCs (Indicators of Compromise).
- **Expand Telemetry Sources:** Incorporate network-level visibility by integrating Suricata or Zeek logs into the Wazuh manager to detect anomalous lateral movement.
- **Continuous Rule Tuning:** Regularly audit and tune detection rules to reduce false positives and align alerting mechanisms closer to the MITRE ATT&CK framework.

Github Repository

Repository Link: <https://github.com/Princymistry/SOC-Operations-Threat-Detection-Mastery>

Conclusion

This project successfully demonstrated the design, deployment, and operationalization of a functional Security Operations Center (SOC) environment using Wazuh SIEM. Throughout the four-week engagement, a centralized security monitoring infrastructure was established, enabling real-time collection, analysis, and correlation of security events from both Windows and Linux endpoints.

The implementation validated core SOC capabilities, including endpoint onboarding, log parsing, threat detection, adversary simulation, and incident visualization. Simulated attacks such as SSH brute-force attempts, malware execution, and privilege escalation activities were effectively detected and

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correlated by the SIEM platform, confirming the effectiveness of the deployed security controls.

Furthermore, the development of custom dashboards transformed raw security telemetry into actionable intelligence, improving operational visibility and supporting informed decision-making. Overall, this project strengthened practical skills in SIEM management, threat hunting, incident analysis, and SOC operations while emphasizing the importance of continuous monitoring, rule tuning, and proactive defense strategies in modern cybersecurity environments.